Peipei Zhou

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Google Scholar: https://scholar.google.com/citations?user=px_jwFgAAAAJ&hl=en

RESEARCH INTEREST

- Customized computer architecture, programming abstraction, and electronic design automation for applications including healthcare, e.g., precision medicine, and artificial intelligence
- Characterization, modeling, and optimization in performance, energy, carbon, and cost for reconfigurable and heterogeneous systems from edge to data center
- Sustainable computing and chiplet-based system

ACADEMIC POSITION

Brown University 2024/09 - Present
Tenure-Track Assistant Professor, School of Engineering

University of Pittsburgh 2021/09 - 2024/08
Tenure-Track Assistant Professor, Department of Electrical and Computer Engineering

EDUCATION

University of California, Los Angeles 2014/06 - 2019/08
Ph.D., Computer Science, GPA: 4.0/4.0 Advisor: Prof. Jason Cong
Dissertation: Modeling and Optimization for Customized Computing: Performance, Energy and Cost Perspective.

University of California, Los Angeles 2012/09 - 2014/06
M.S., Electrical Engineering, GPA: 3.8/4.0 Advisor: Prof. Jason Cong
Thesis: A Fully Pipelined and Dynamically Composable Architecture of CGRA.

Southeast University, Chien-Shiung Wu Honors College 2008/08 - 2012/06
B.S., Electrical and Computer Engineering, GPA: 3.9/4.0, Rank: 1/90

HONORS AND AWARDS

- **Selected for the National Academy of Engineering's Grainger Foundation Frontiers of Engineering Symposium, 2026.** Recognizing exceptional research and technical leadership among early-career engineers.
(Brown_News, NAE_News)
- **National Science Foundation Faculty Early Career Development (CAREER) Award, 2026** (News)

[1]Updated October 12, 2026

- **IEEE/ACM International Conference on Computer-Aided Design (ICCAD) 10-Year Retrospective Most Influential Paper Award, 2025**, "Caffeine: Towards Uniformed Representation and Acceleration for Deep Convolutional Neural Networks", 2025 William J. McCalla ICCAD 10-Year Retrospective Most Influential Paper Award (ICCAD 2025) (News)
- **ACM IGSC Best Paper Award, 2026**, "Advancing Environmental Sustainability in Data Centers via Carbon Depreciation Models", 2026 16th IEEE International Green and Sustainable Computing (IGSC 2026)
- **ACM/SIGDA FPGA Best Paper Nominee, 2025**, "ARIES: An Agile MLIR-Based Compilation Flow for Reconfigurable Devices with AI Engines", 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA_2025)
- **IEEE IGSC Best Viewpoint Paper, 2024**, "Amortizing Embodied Carbon Across Generations", 2024 15th IEEE International Green and Sustainable Computing (IGSC 2024)
- **IEEE/ACM IGSC Best Viewpoint Paper Finalist, 2023**, "REFRESH FPGAs: Sustainable FPGA Chiplet Architectures", 2023 IEEE/ACM International Green and Sustainable Computing (IGSC 2023)
- **IEEE Council on Electronic Design (CEDA) IEEE Transactions on Computer-Aided Design Donald O. Pederson Best Paper Award, 2019**, "Caffeine: Towards Uniformed Representation and Acceleration for Deep Convolutional Neural Networks", 2 out of 1000+ submissions, awarded annually to recognize the best paper published in the IEEE Transactions on CAD in the two calendar years preceding the award. IEEE CEDA Award Link and UCLA Computer Science Department Award Link
- **IEEE/ACM ICCAD Best Paper Nominee, 2018**, "SODA: Stencil with Optimized Dataflow Architecture", 6 papers out of 400 submissions, 2018 IEEE/ACM International Conference on Computer-Aided Design (ICCAD 2018)
- **IEEE ISPASS Best Paper Nominee, 2018**, "Doppio: I/O-Aware Performance Analysis, Modeling and Optimization for In-Memory Computing Framework", 4 papers out of 67 submissions, 2018 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS 2018) ISPASS 2018 Program Link
- **UCLA Outstanding Recognition in Research in Computer Science**, awarded annually to recognize top 3 outstanding Ph.D. Researcher in UCLA CS department, 2019.

  UCLA Samueli School of Engineering Award Link and UCLA VAST Lab Award Link
- **Phi Tau Phi Scholarship**, awarded annually in recognition of academic achievements and scholarly contributions in West America, 2018

  Phi Tau Phi Scholastic Honor Society of America Award Link
- **Best Poster Award**, High Throughput Sequencing (HiTSeq 2015)
- **Honeywell Innovator Scholarship**, one of five recipients in Mainland China, 2011

PEER-REVIEWED PUBLICATIONS

Summary: 43 peer-reviewed publications, including 32 published since the start of my independent academic career in 2021 and 17 published since joining Brown University in 2024. My name is underlined. Ph.D. students advised by me are shown in **boldface**.

43. [**DAC'26**] **Xingzhen Chen**, **Jinming Zhuang**, **Zhuoping Yang**, **Shixin Ji**, **Sarah Schultz**, Zheng Dong, Weisong Shi, Peipei Zhou, "FILCO: Flexible Composing Architecture with Real-Time Reconfigurability for DNN Acceleration," 63rd ACM/IEEE Design Automation Conference (DAC'26), acceptance ratio: 454/2434 = 18.6%, July 26 - July 29, 2026, Long Beach, CA, USA.

42. [**DAC'26**] **Shixin Ji**, **Jinming Zhuang**, **Sarah Schultz**, **Zhuoping Yang**, **Xingzhen Chen**, Zheng Dong, Alex K. Jones, Yihui Ren, Peipei Zhou, "PHAROS: Pipelined Heterogeneous Accelerators for Real-time Safety-critical Systems With Deadline Compliance," 63rd ACM/IEEE Design Automation Conference (DAC'26), acceptance ratio: 454/2434 = 18.6%, July 26 - July 29, 2026, Long Beach, CA, USA.

41. [**GLSVLSI'26**] **Shixin Ji**, **Jinming Zhuang**, **Zhuoping Yang**, **Xingzhen Chen**, **Wei Zhang**, Peipei Zhou, "μ-ORCA: Optimizing Acceleration for Microsecond-Scale Deep Neural Network Inference on ACAP," 36th ACM Great Lakes Symposium on VLSI (GLSVLSI'26), June 22-24, 2026, Canandaigua, NY, USA.

40. [**GLSVLSI'26**] **Xingzhen Chen**, **Zhuoping Yang**, **Jinming Zhuang**, **Shixin Ji**, **Sarah Schultz**, Zheng Dong, Weisong Shi, Peipei Zhou, "DORA: Dataflow-Instruction Orchestration Architecture for DNN Acceleration," 36th ACM Great Lakes Symposium on VLSI (GLSVLSI'26), June 22-24, 2026, Canandaigua, NY, USA.

39. [**IGSC'26**] **Shixin Ji**, **Zhuoping Yang**, **Xingzhen Chen**, Alex Jones, Peipei Zhou, "Advancing Environmental Sustainability in Data Centers via Carbon Depreciation Models," The 16th International Green and Sustainable Computing Conference (IGSC'26), June 22-24, 2026, Canandaigua, NY, USA.
(IGSC'26 Best Paper Award)

38. [**IGSC'26**] **Xingzhen Chen**, **Shixin Ji**, Zheng Dong, Peipei Zhou, "To Overlay or to Customize? Revisiting Architectural Choices in Heterogeneous Systems," The 16th International Green and Sustainable Computing Conference (IGSC'26), June 22-24, 2026, Canandaigua, NY, USA.

37. [**JCST'26**] Weisong Shi, Zheng Dong, Peipei Zhou, "Physical Intelligence on the Edge: A Vision for the Decade Ahead," Journal of Computer Science and Technology, 2026.

36. [**RTSS'25**] **Shixin Ji**, **Zhuoping Yang**, **Xingzhen Chen**, **Wei Zhang**, **Jinming Zhuang**, Alex Jones, Zheng Dong, Peipei Zhou, "DERCA: DetERministic Cycle-Level Accelerator on Reconfigurable Platforms in DNN-Enabled Real-Time Safety-Critical Systems," Proceedings of the 46th IEEE Real-Time Systems Symposium, RTSS 2025, December 2-5, 2025, Boston, MA, USA. Open-source: https://github.com/arc-research-lab/DERCA.

35. [**SC'25**] **Zhuoping Yang**, **Jinming Zhuang**, **Xingzhen Chen**, Alex Jones, Peipei Zhou, "AGILE: Lightweight and Efficient Asynchronous GPU-SSD Integration," Proceedings of the

2025 ACM International Conference for High Performance Computing, Networking, Storage, and Analysis, SC 2025, Nov. 16 - Nov. 21, 2025, St. Louis, MO, USA. Open-source: https://github.com/arc-research-lab/AGILE.

34. [**GLSVLSI'25**] **Shixin Ji**, **Xingzhen Chen**, **Jinming Zhuang**, **Wei Zhang**, **Zhuoping Yang**, **Sarah Schultz**, Yukai Song, Jingtong Hu, Alex Jones, Zheng Dong, Peipei Zhou, "ART: Customizing Accelerators for DNN-Enabled Real-Time Safety-Critical Systems," Proceedings of the 2025 ACM Proceedings of the Great Lakes Symposium on VLSI 2025, 30 June 2025 - 2 July 2025, New Orleans, LA, USA.

33. [**TECS'25**] Reza Jafarpourmarzouni, FNU Sumaiya, Ruoxiang Li, Nan Guan, Guang Wang, Peipei Zhou, Zheng Dong, "Reaction Latency Analysis of Message Synchronization in Edge-assisted Autonomous Driving," ACM Transactions on Embedded Computing Systems, 2025.

32. [**FPGA'25**] **Jinming Zhuang**, Shaojie Xiang, Hongzheng Chen, Niansong Zhang, **Zhuoping Yang**, Tony Mao, Zhiru Zhang, Peipei Zhou, "ARIES: An Agile MLIR-Based Compilation Flow for Reconfigurable Devices with AI Engines," Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA'25), Feb 27 - Mar 1, 2025, Monterey, CA, USA. Open-source: https://github.com/arc-research-lab/Aries. **(FPGA'25 Best Paper Nominee)**.

31. [**TCAD'25**] Yue Tang, Alex K. Jones, Jinjun Xiong, Peipei Zhou, Jingtong Hu, "MTrain: Enable Efficient CNN Training on Heterogeneous FPGA-based Edge Servers," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2025.

30. [**TCAD'24**] Peiyan Dong*, **Jinming Zhuang***, **Zhuoping Yang**, **Shixin Ji**, Dongkuan Xu, Yanyu Li, Heng Huang, Jingtong Hu, Alex K. Jones, Yiyu Shi, Yanzhi Wang, Peipei Zhou, "EQ-ViT: Algorithm-Hardware Co-Design for End-to-End Acceleration of Real-Time Vision Transformer Inference on Versal ACAP Architecture," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, November, 2024. (* co-first authors)

29. [**TCAD'24**] Yue Tang, Yukai Song, Naveena Elango, Sheena Ratnam Priya, Alex K. Jones, Jinjun Xiong, Peipei Zhou, Jingtong Hu, "CHEF: A Framework for Deploying Heterogeneous Models on Clusters with Heterogeneous FPGAs," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, November.

28. [**ESWEEK'24**] **Zhuoping Yang**, **Wei Zhang**, **Shixin Ji**, Peipei Zhou, Alex K Jones, "Reducing Smart Phone Environmental Footprints with In-Memory Processing," 2024 International Conference on Hardware/Software Codesign and System Synthesis, November 2024 (CODES+ISSS 2024 in ESWEEK).

27. [**IGSC'24**] **Shixin Ji**, **Jinming Zhuang**, **Zhuoping Yang**, Alex K Jones, Peipei Zhou, "Amortizing Embodied Carbon Across Generations," 15th International Green and Sustainable Computing Conference (IGSC'24).
(IEEE IGSC'24 Best Viewpoint Paper).

Note: The publications listed above were papers published since joining Brown University in 2024.

26. [**TRETS'24**] **Jinming Zhuang**, Jason Lau, Hanchen Ye, **Zhuoping Yang**, **Shixin Ji**, Jack Lo, Kristof Denolf, Stephen Neuendorffer, Alex Jones, Jingtong Hu, Yiyu Shi, Deming Chen, Jason Cong, Peipei Zhou, "CHARM 2.0: Composing Heterogeneous Accelerators for Deep Learning on Versal ACAP Architecture," Accepted at ACM Transactions on Reconfigurable Technology and Systems, September 2024. Open-source: https://github.com/arc-research-lab/CHARM.

25. [**TRETS'24**] Yuqi Li, Kehao Zhao, Jieru Zhao, Qirui Wang, Shuda Zhong, Nageswara Lalam, Ruishu Wright, Peipei Zhou, Kevin P. Chen, "FiberFlex: Real-time FPGA-based Intelligent & Distributed Fiber Sensor System for Pedestrian Recognition," Accepted at ACM Transactions on Reconfigurable Technology and Systems, November 2024.

24. [**ISVLSI'24**] **Shixin Ji**, **Zhuoping Yang**, **Xingzhen Chen**, Stephen Cahoon, Jingtong Hu, Yiyu Shi, Alex Jones, Peipei Zhou, "SCARIF: Towards Carbon Modeling of Cloud Servers with Accelerators," 2024 IEEE Computer Society Annual Symposium on VLSI, ISVLSI 2024. Open-source: https://github.com/arc-research-lab/SCARIF.

23. [**DAC'24**] Ruiyang Qin, Jun Xia, Zhenge Jia, Meng Jiang, Ahmed Abbasi, Peipei Zhou, Jingtong Hu, Yiyu Shi, "Enabling On-Device Large Language Model Personalization with Self-Supervised Data Selection and Synthesis," 61st ACM/IEEE Design Automation Conference, San Francisco, California, USA, (DAC'24).

22. [**FPGA'24**] **Jinming Zhuang**, **Zhuoping Yang**, **Shixin Ji**, Heng Huang, Alex K. Jones, Jingtong Hu, Yiyu Shi, Peipei Zhou, "SSR: Spatial Sequential Hybrid Architecture for Latency Throughput Tradeoff Design Space Exploration," Proceedings of the 2024 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA'24), March 3-5, 2024, Monterey, CA, USA. Open-source: https://github.com/arc-research-lab/SSR.

21. [**ASPDAC'24**] **Zhuoping Yang**, **Shixin Ji**, **Xingzhen Chen**, **Jinming Zhuang**, Weifeng Zhang, Dharmesh Jani, Peipei Zhou, "Challenges and Opportunities to Enable Large-Scale Computing via Heterogeneous Chiplets," Proceedings of the 28th Asia and South Pacific Design Automation Conference, ASPDAC'24, Incheon Songdo Convensia, South Korea.

20. [**IGSC'23**] Peipei Zhou, **Jinming Zhuang**, Stephen Cahoon, Yue Tang, **Zhuoping Yang**, **Xingzhen Chen**, Yiyu Shi, Jingtong Hu, Alex K. Jones, "REFRESH FPGAs: Sustainable FPGA Chiplet Architectures," 14th International Green and Sustainable Computing Conference (IGSC'23).
(IGSC'23 Best Viewpoint Paper Finalist)

19. [**ICCAD'23**] **Zhuoping Yang**, **Jinming Zhuang**, Jiaqi Yin, Cunxi Yu, Alex K. Jones, Peipei Zhou, "AIM: Accelerating Arbitrary-precision Integer Multiplication on Heterogeneous Reconfigurable Computing Platform Versal ACAP," IEEE/ACM International Conference on Computer-Aided Design, (ICCAD'23), acceptance ratio: 21%, October 29 - November 2, 2023, San Francisco, CA, USA. Open-source: https://github.com/arc-research-lab/AIM.

18. [**DAC'23**] **Jinming Zhuang**, **Zhuoping Yang**, Peipei Zhou, "High Performance, Low Power Matrix Multiply Design on ACAP: from Architecture, Design Challenges, and DSE

Perspectives," 60th ACM/IEEE Design Automation Conference, San Francisco, California, USA, (DAC'23), acceptance ratio: 263/1157 = 22.7%, July 9-13, 2023, San Francisco, CA, USA. Open-source: https://github.com/arc-research-lab/CHARM.

17. [**FPGA'23**] **Jinming Zhuang**, Jason Lau, Hanchen Ye, **Zhuoping Yang**, Yubo Du, Jack Lo, Kristof Denolf, Stephen Neuendorffer, Alex Jones, Jingtong Hu, Deming Chen, Jason Cong, Peipei Zhou, "CHARM: Composing Heterogeneous Accelerators for Matrix Multiply on Versal ACAP Architecture," Proceedings of the 2023 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '23), February 12-14, 2023, Monterey, CA, USA. Open-source: https://github.com/arc-research-lab/CHARM.
No.1 Downloaded and Cited Paper in 2023 ACM/SIGDA FPGA Proceedings

16. [**IEEE Micro'23**] Sebastien Ollivier, Sheng Li, Yue Tang, Stephen Cahoon, Ryan Caginalp, Chayanika Chaudhuri, Peipei Zhou, Xulong Tang, Jingtong Hu, Alex K. Jones, "Sustainable AI Processing at the Edge," IEEE Micro.

15. [**TCAD'22**] Yue Tang, Yawen Wu, Peipei Zhou, and Jingtong Hu, "Enabling Weakly Supervised Temporal Action Localization From On-Device Learning of the Video Stream," in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, vol. 41, no. 11, pp. 3910-3921, Nov. 2022, doi: 10.1109/TCAD.2022.3197536.

14. [**DAC'22**] Xinyi Zhang, Cong Hao, Peipei Zhou, Alex Jones, Jingtong Hu. "H2H: Heterogeneous Model to Heterogeneous System Mapping with Computation and Communication Awareness". 59th ACM/IEEE Annual Design Automation Conference 2022 (DAC '22).

13. [**TODAES'22**] Yue Tang, Xinyi Zhang, Peipei Zhou, Jingtong Hu. "EF-Train: Enable Efficient On-device CNN Training on FPGA Through Data Reshaping for Online Adaptation or Personalization". ACM Transactions on Design Automation of Embedded Systems (TODAES) Special Issue on Energy-Efficient AI Chips (TODAES '22).
Student Yue won TODAES Rookie Author of the Year (RAY) Award 2023

12. [**TECS'21**] Xinyi Zhang, Yawen Wu, Peipei Zhou, Xulong Tang, Jingtong Hu. "Algorithm-hardware Co-design of Attention Mechanism on FPGA Devices". ACM Transactions on Embedded Computing Systems (TECS '21).

Note: The publications listed above were published since the start of my independent academic career in 2021.

11. [**FPGA'21**] Peipei Zhou*, Jiayi Sheng, Cody Hao Yu, Peng Wei, Jie Wang, Di Wu, Jason Cong. "MOCHA: Multinode Cost Optimization in Heterogeneous Clouds with Accelerators". 2021 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA).

10. [**FCCM'20**] Michael Lo, Zhenman Fang, Jie Wang, Peipei Zhou, Mau-Chung Frank Chang, Jason Cong. "Algorithm-Hardware Co-design for BQSR Acceleration in Genome Analysis ToolKit". IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM), May 2020

9. [**ICCAD'18**] Yuze Chi, Jason Cong, Peng Wei, Peipei Zhou. "SODA: Stencil with Optimized Dataflow Architecture". IEEE/ACM International Conference on Computer-Aided Design

(ICCAD), November 2018, best paper nominee ratio: 6/400 = 1.5%
(ICCAD'18 Best Paper Nominee)

8. [**TCAD'18**] Chen Zhang, Guangyu Sun, Zhenman Fang, Peipei Zhou, Peichen Pan, Jason Cong. "Caffeine: Towards Uniformed Representation and Acceleration for Deep Convolutional Neural Networks". IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), October 2018
(IEEE TCAD 2019 Donald O. Pederson Best Paper Award)

7. [**FCCM'18**] Jason Cong, Peng Wei, Cody Hao Yu, Peipei Zhou*. "Latte: Locality Aware Transformation for High-Level Synthesis". IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM), May 2018, acceptance ratio: 7/48 = 14.6%

6. [**FCCM'18**] Zhenyuan Ruan, Tong He, Bojie Li, Peipei Zhou, Jason Cong. "ST-Accel: A High-Level Programming Platform for Streaming Applications on FPGA". IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM), May 2018, acceptance ratio: 22/106 = 20.7%

5. [**ISPASS'18**] Peipei Zhou*, Zhenyuan Ruan, Zhenman Fang, Megan Shand, David Roazen, Jason Cong Doppio: I/O-Aware Performance Analysis, Modeling and Optimization for In-Memory Computing Framework. IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), April 2018, best paper nominee ratio: 4/67 = 5.9%
(ISPASS'18 Best Paper Nominee)

4. [**DAC'17**] Jason Cong, Peng Wei, Cody Hao Yu, Peipei Zhou. "Bandwidth Optimization Through On-Chip Memory Restructuring for HLS" 54th Annual Design Automation Conference (DAC), June 2017, acceptance rate: 161/676 = 24%

3. [**ICCAD'16**] Chen Zhang, Zhenman Fang, Peipei Zhou, Peichen Pan, Jason Cong. "Caffeine: Towards Uniformed Representation and Acceleration for Deep Convolutional Neural Networks". IEEE/ACM International Conference on Computer-Aided Design (ICCAD), November 2016, acceptance rate: 97/408 = 23.8%

2. [**FCCM'16**] Peipei Zhou*, Hyunseok Park, Zhenman Fang, Jason Cong, André DeHon. "Energy Efficiency of Full Pipelining: A Case Study for Matrix Multiplication". IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM), May 2016, acceptance rate: 32/133 = 24.1%

1. [**FCCM'14**] Jason Cong, Hui Huang*, Chiyuan Ma, Bingjun Xiao*, Peipei Zhou*. "A Fully Pipelined and Dynamically Composable Architecture of CGRA". IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM), May 2014, acceptance rate: 22/134 = 16.4%

Note: The publications with an asterisk (*) next to my name were first-author publications during my Ph.D. studies.

BOOK CHAPTERS

2. **Shixin Ji**, **Xingzhen Chen**, **Jinming Zhuang**, **Zhuoping Yang**, **Sarah Schultz**, Zheng Dong, Weisong Shi, Peipei Zhou, "Architecting Safety-Aware DNN Accelerators for Real-Time Physical Intelligence Systems." In Handbook of Embedded Machine Learning: Cyber-Physical, IoT, and Edge Computing, Springer Nature, forthcoming, expected publication: 2026.

1. Yawen Wu, Yue Tang, Dewen Zeng, Xinyi Zhang, Peipei Zhou, Yiyu Shi, and Jingtong Hu. "Efficient hardware and software design for on-device learning." In Embedded Machine Learning for Cyber-Physical, IoT, and Edge Computing: Software Optimizations and Hardware/Software Codesign, pp. 371-404. Cham: Springer Nature Switzerland, 2023.

PREPRINTS AND TECHNICAL REPORTS

5. [**arXiv'26**] Peipei Zhou, Zheng Dong, Insup Lee, Aidong Zhang, Robert Dick, Majid Sarrafzadeh, Xiaodong Wu, Weisong Shi, Zhuoping Yang, Jingtong Hu, Yiyu Shi. "Report for NSF Workshop on Algorithm-Hardware Co-design for Medical Applications." arXiv preprint arXiv:2603.10976 (2026).

4. [**arXiv'18**] Jason Cong, Zhenman Fang, Yuchen Hao, Peng Wei, Cody Hao Yu, Chen Zhang, Peipei Zhou. "Best-Effort FPGA Programming: A Few Steps Can Go a Long Way". arXiv:1807.01340 [cs.AR], July 2018

3. [**FPGA'18**] Yuze Chi, Peipei Zhou, Jason Cong. "An Optimal Microarchitecture for Stencil Computation with Data Reuse and Fine-Grained Parallelism (Abstract Only)". ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA), February 2018

2. [**FPGA'16**] Yu-Ting Chen, Jason Cong, Zhenman Fang, Peipei Zhou. "ARAPrototyper: Enabling Rapid Prototyping and Evaluation for Accelerator-Rich Architecture (Abstract Only)". IEEE ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA), February 2016

1. [**HitSeq'16**] Yu-Ting Chen, Jason Cong, Jie Lei, Sen Li, Myron Peto, Paul Spellman, Peng Wei, and Peipei Zhou. "CS-BWAMEM: A fast and scalable read aligner at the cloud scale for whole genome sequencing (Abstract Only)" High Throughput Sequencing, Algorithms,& Applications (HiTSeq), an ISMB/ECCB 2015 special interest group (SIG) satellite conference, July 2015

STUDENT ADVISING

- Doctoral Advisor for six current Ph.D. students and one Ph.D. graduate:

Ph.D. Graduate:
Jinming Zhuang (Ph.D., August 2026; first Ph.D. advisee; joined the Department of Computer Engineering at Rochester Institute of Technology as a tenure-track Assistant Professor)
Current Ph.D. Students:
Zhuoping Yang (2022/08–Present)
Shixin Ji (2023/08–Present)
Xingzhen Chen (2023/08–Present)

Sarah Schultz (2024/08–Present)
Wei Zhang (2024/08–Present)
Jack Adiletta (2026/09–Present)

- Undergraduate Honors Thesis Advisor for two students:
  Gannon Lemaster '26 (2025/09–2026/05, Honors Thesis), entered the Ph.D. program at MIT
  Devon Kear-Leng '26 (2025/09–2026/05, Honors Thesis), joined Apex in summer 2026
- Undergraduate Research Advisor for two students:
  Daniel Yang '29 (2025/12–Present)
  Wilson Vo '26 (2025/09–2026/05), admitted to the M.S. program at CMU
- Master Thesis Advisor for one student:
  Ye Yuan (2026/04–Present)
- Doctoral Exam Reader at Brown:
  Zack Adamson (2025/08–Present)
  Tasneem Shaffee (2026/05–Present)
- Doctoral Dissertation Committee Member at Pitt:
  Xin Xin (2021/07–2023/06, now tenure-track faculty at University of Central Florida)
  Yuqi Li (2022/02–2024/11)
  Yue Tang (2022/04–2024/07)
  Yanan Guo (2022/11–2024/06, now tenure-track faculty at University of Rochester)
  Noah Perryman (2022/11–2024/07)
  Luke Kljucaric (2022/12–2023/10)
  Tyler Garrett (2022/12–2024/07)
  Jieru Zhao (2023/04–2024/07)
  Calvin Gealy (2023/08–2024/12)

STUDENT AWARDS & PLACEMENT

- Jinming Zhuang (Ph.D., August 2026): Appointed as Tenure-Track Assistant Professor, Computer Engineering Department, Rochester Institute of Technology, in August 2026.
- Shixin Ji (Ph.D. expected August 2028): ACM/IEEE DAC Young Fellow, 2026.
- Xingzhen Chen (Ph.D. expected August 2028): ACM/IEEE DAC Young Fellow, 2026.
- Sarah Schultz (Ph.D. expected August 2029): Jan Tauc Fellowship in Engineering, Brown University (2025–2026).
- Zhuoping Yang (Ph.D. expected August 2027): ACM GLSVLSI Service Recognition Award (2026).
- Gannon Lemaster '26: National Science Foundation Graduate Research Fellowship Program (NSF GRFP), 2026; will begin Ph.D. studies at MIT in Fall 2026.
- Wilson Vo '26: will begin graduate study at Carnegie Mellon University, Fall 2026.
- Devon Kear-Leng '26: Joined Apex - Satellite Platforms, Summer 2026.

COURSES TAUGHT

Courses taught at Brown University

- 2024-2025 Fall: ENGN2911X Reconfigurable Computing (Graduate Course, attendees: 24)
- 2024-2025 Spring: ENGN1600 Design and Implementation of Digital Integrated Circuits (Undergraduate Course, attendees: 32)
- 2025-2026 Fall: ENGN2911X Reconfigurable Computing (Graduate Course, attendees: 24)

Courses taught at University of Pittsburgh

- 2021-2022 Spring: ECE2195 Reconfigurable Computing in Deep Learning (Graduate Course, attendees: 23), Teaching Score: 4.10 / 5.00
- 2022-2023 Fall: ECE1110 Computer Organization and Architecture (Undergraduate Course, attendees: 51), Teaching Score: 4.34 / 5.00
- 2022-2023 Spring: ECE2195 Reconfigurable Computing in Deep Learning (Graduate Course, attendees: 14), Teaching Score: 4.48 / 5.00
- 2022-2023 Spring: ECE1175 Embedded Systems Design (Undergraduate Course, attendees: 17), Teaching Score: 4.50 / 5.00
- 2023-2024 Fall: ECE1110 Computer Organization and Architecture (Undergraduate Course, attendees: 60), Teaching Score: 4.16 / 5.00
- 2023-2024 Spring: ECE1110 Computer Organization and Architecture (Undergraduate Course, attendees: 17), Teaching Score: 4.78 / 5.00

TALKS

- **Summary**: 25+ Invited talks at leading universities (University of Rhode Island, University of Delaware, University of North Texas, Wayne State University, Rutgers University, University of Connecticut, Temple University, Texas A&M University, Duke University, and Carnegie Mellon University), national laboratories (Brookhaven National Laboratory), industry venues (GE Vernova, Mathworks), national cyberinfrastructure programs (NSF FABRIC), international conferences and workshops (FCCM-CASI, USENIX VCEW, ISVLSI, ESWEEK, DAC-ROAD4NN, and IWLS), professional society events (IEEE Day Tech), and major community organizations including the Open Compute Project (OCP).
- 2026/11, University of Rhode Island, Department of Electrical, Computer, and Biomedical Engineering Seminar, Kingston, RI, Invited Talk: "Efficient Design and Programming of Heterogeneous Reconfigurable Computing for AI" (Scheduled)
- 2026/09/29, GE Vernova AI-EDGE Symposium 2026, Niskayuna, NY, Invited Talk: "Efficient Design and Programming of Heterogeneous Reconfigurable Computing for AI"
- 2026/09/03, Worcester Polytechnic Institute (WPI), Department of Electrical and Computer Engineering Seminar, Worcester, MA, Invited Talk: "Efficient Programming on Heterogeneous Accelerators"
- 2026/08, NSF FABRIC (National Research Infrastructure) Webinar Series, Invited Talk: "AGILE: Lightweight and Efficient Asynchronous GPU-SSD Integration" (Scheduled)
- 2026/05/13, CASI: Workshop on Custom Architectures for Scientific Instruments at IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM 2026), Atlanta, GA, Invited Talk: "Architecting Hardware/Software Co-Design on Heterogeneous Reconfigurable Computing"

- 2026/05/08, Initiative for Sustainable Energy (ISE) Mini-Symposium, Providence, RI, Invited Talk: "Characterization and Optimization of Heterogeneous Computing for AI"
  EventPhotos
- 2025/10/14, NSF FABRIC KNIT 11 Community Workshop, Salt Lake City, UT, Invited Talk: "SPECTRUM: Architecturing Evolutionary Rapid-Prototyping Testbed for Low Latency Multi-Domain Computing"
- 2025/10/07, Distinguished Speaker for the IEEE Lafayette Section Young Professionals Affinity Group "IEEE Day Tech Chat" event, Invited Talk: "Efficient Programming on Heterogeneous Accelerators"
- 2025/09/12, Brookhaven National Laboratory New York Scientific Data Summit 2025: Powering the Future of Science with Artificial Intelligence, New York, NY, Invited Talk: "Efficient Programming on Heterogeneous Accelerators: Architecture and Compiler Insights"
- 2025/06/17, MathWorks Research Summit 2025, Natick, MA, Invited Talk: "Energy Efficient AI and Computation"
- 2025/06/09, USENIX Vail Computer Elements Workshop (VCEW) 2025, Invited Talk: "Efficient Programming on Heterogeneous Accelerators"
- 2025/05/09, University of Delaware, Newark, DE, Computer & Information Sciences Seminar, Invited Talk: "Efficient Programming on Heterogeneous Accelerators"
  Recorded Talk on YouTube: https://www.youtube.com/watch?v=1Wo4FOTsTsM
- 2025/02/06, University of North Texas, Data Science Talk Series, Invited Talk: "Efficient Programming on Heterogeneous Accelerators"
- 2024/12/03, Wayne State University, Computer Science Department Seminar, Detroit, MI, Invited Talk: "Efficient Programming on Heterogeneous Accelerators"
- 2024/11/26, Rutgers Efficient AI (REFAI) Seminar, Rutgers University, Invited Talk: "Efficient Programming on Heterogeneous Accelerators"
- 2024/10/25, ECE Colloquium, University of Connecticut, Storrs, CT, Invited Talk: "Efficient Programming on Heterogeneous Accelerators"
- 2024/07/03, ISVLSI 2024 Special Session on Sustainable Computing from Edge to Data Center, Knoxville, TN, Invited Talk: "SCARIF: Towards Carbon Modeling of Cloud Servers with Accelerators"
  Slides: https://peipeizhou-eecs.github.io/publication/2024_isvlsi_scarif/
- 2024/04/19, Open Compute Project (OCP) AI-HW-SW-CoDesign Working Group, Invited Talk: "Efficient Programming on Heterogeneous Accelerators"
- 2023/10/18, San Jose Convention Center, 2023 Open Compute Project (OCP) Global Summit, Future Technology Symposium, Invited Talk: "Architectural Challenges and Innovation for Compute Infrastructure Co-Design"
  Recorded Talk on YouTube: https://www.youtube.com/watch?v=Lp5Al9vbqLg
  Slides: https://peipeizhou-eecs.github.io/talk/architectural-challenges-and-innovation-for-compute-infrastructure-co-design/OCP23G_SYM-PeipeiZhou.pdf
- 2023/08/30, Temple University, Department of Electrical and Computer Engineering Seminars, Invited Talk: "CHARM: Composing Heterogeneous Accelerators for Matrix Multiply on

Versal ACAP Architecture"
https://events.temple.edu/fall-seminar-series

- 2023/04/21, TAMU CESG Seminars, Invited Talk: "CHARM: Composing Heterogeneous Accelerators for Matrix Multiply on Versal ACAP Architecture"
https://cesg.tamu.edu/2023/03/01/cesg-seminar-peipei-zhou/
- 2023/04/07, CMU Crossroads Seminars, Pittsburgh, PA, Invited Talk: "CHARM: Composing Heterogeneous Accelerators for Matrix Multiply on Versal ACAP Architecture"
YouTube: https://youtu.be/Hl__s5WqSZA
Slides: https://peipeizhou-eecs.github.io/publication/fpga23/FPGA2023_CHARM.pdf
- 2022/10, 18th EMBEDDED SYSTEMS WEEK, Invited Tutorial: "Practice on Performance Autotuning in AI Compute Chip"
https://esweek.org/wp-content/uploads/2022/10/ESWEEK_2022_booklet.pdf
- 2022/7, DAC-ROAD4NN 2022 International Workshop on Research Open Automatic Design for Neural Networks 2022 Invited Talk: "Practice on Performance Autotuning in AI Compute Chip"
- 2022/7, 31st International Workshop on Logic & Synthesis, **Keynote**: "Practice on Performance Autotuning in AI Compute Chip",
https://www.iwls.org/iwls2022/
- 2022/1, The Pittsburgh Center for AI Innovation in Medical Imaging (CAIIMI) SEMINAR: "Domain-Specific Computing for Artificial Intelligence",
https://twitter.com/PittCAIIMI/status/1483443239344652291
- 2021/10, Duke ECE SEMINAR: "Practice on Performance Autotuning in AI Compute Chip",
https://ece.duke.edu/about/events/3210
- 2020/10, IEEE Pittsburgh Section: "Women in ECE Panel Discussion",
https://events.vtools.ieee.org/m/241997
- 2018/12, "Customizable Domain-Specific Computing (3-Minute Lightning Talk)". National Science Foundation (NSF) 10-Year Expeditions Anniversary PI Meeting: 10 Years of Transforming Science & Society, Washington. D.C.
- 2019/02, "Mocha: Multinode Optimization of Cost in Heterogeneous Cloud with Accelerators (Technology Review Talk)". Falcon Computing Solutions Technology Review, Los Angeles
- 2018/11, "Cost Optimization Engine for Heterogeneous Public Cloud Featuring Genomics Workloads (Technology Review Talk)". Falcon Computing Solutions Technology Review, Los Angeles
- 2017/06, "Analyzing and Accelerating Genome Analysis Toolkit GATK4". CDSC/InTrans Project Annual Review, Los Angeles. The Center for Domain-Specific Computing (CDSC) was established in 2009. The initial funding for CDSC was provided by the National Science Foundation under the Expedition in Computing Program. Intel Corporation becomes the first industrial partner to provide financial support to CDSC under the Innovation Transitions (InTrans) Program. Participating universities are the University of California, Los Angeles, Rice University, Ohio State University, Oregon Health, and Science University (OHSU).
- 2016/10, "Quantifying the I/O Impact on Apache Spark: with Application to Computational Genomic". CDSC/InTrans Project Annual Review, Los Angeles.

- 2016/10, "Fully Pipeline or Not? An Energy Perspective". CDSC/InTrans Project Annual Review, Los Angeles
- 2016/01, "Parallelization of CAVIAR and Identification of Causal Variants in Breast Cancer". CDSC/InTrans Project Annual Review, Los Angeles.
- 2014/05, "Prototype of A Fully Pipelined Configurable Array FPCA". CDSC Year 5 Semi-Annual Review, Los Angeles.
- 2013/10, "FPGA Prototyping of Accelerator-Rich Architectures". CDSC Year 4 Semi-Annual Review, Los Angeles.

DEMOS

- 2026/05/14, Demo at IEEE In- ternational Symposium on Field-Programmable Custom Computing Machines (FCCM 2026)
  Demo 1: Towards μs-Scale DNN Inference on Versal ACAP
  Demo 2: FILCO: Flexible Composing Architecture with Real-Time Reconfigurability on Versal Platform
  EventPhotos
- 2025/09/25, Demo at Rhode Island Senate and Rhode Island House of Representatives and Civic Leaders' Visit at Brown
  EventPhotos
- 2024/10/01, ESWEEK 2024 Embedded Systems Software Competition (ESSC), Demo: "AIM: Accelerating Arbitrary-precision Integer Multiplication on Heterogeneous Reconfigurable Computing Platform Versal ACAP", Demo Presenters: Ph.D. students Zhuoping Yang, Jinming Zhuang, and PI Zhou
  Paper_EventPhotos
- 2024/07/03, ISVLSI 2024 Special Session on Sustainable Computing from Edge to Data Center, Invited Talk: "SCARIF: Towards Carbon Modeling of Cloud Servers with Accelerators"
  Demo_Video_SCARIF_Usage
- 2024/06/25, DAC 2024 University Demonstration, Demo: "Enabling On-Device Large Language Model Personalization with Self-Supervised Data Selection and Synthesis", Demo Presenter: Collaborator Student Ruiyang Qin (Advisor: Prof. Yiyu Shi).
- 2023/09/26, NSF FABRIC KNIT 7, the third in-person FABRIC Community Workshop, Demo: "CHARM: Composing Heterogeneous AcceleRators for End-to-end Deep Learning Inference on Versal ACAP Architecture", Demo Presenters: Ph.D. students Jinming Zhuang, Zhuoping Yang, and PI Zhou
  Poster_EventPhotos
- 2023/07/10, DAC 2023 University Demonstration, Demo: "FPGA HLS acceleration on real-time fiber-optic-based pedestrian recognition system", Demo Presenter: Collaborator Student Yuqi Li (advisor: Prof. Kevin Chen), who I mentored, won 3rd Place Award at DAC 2023 University Demonstration
  News_Certificate_Paper_EventPhotos

- 2023/07/10, DAC 2023 University Demonstration, Demo: "Versal-ViT: A Fully Pipelined Spatial Accelerator Design for Vision Transformers on Versal ACAP", Demo Presenter: Ph.D. student Jinming Zhuang and PI Zhou
  Paper_EventPhotos
- 2022/05/16, FCCM 2022 Demo: "XDSE: A Heterogeneous DSE Framework on Versal Platforms", Demo Presenter: Ph.D. student Jinming Zhuang and PI Zhou
  Poster_EventPhotos

ACADEMIC SERVICES

- **Associate Editor**

  - IEEE Transactions on Computers (Area: Architecture Design and Optimization), 2023-Present

  - ACM Transactions on Reconfigurable Technology and Systems, 2024-Present

  - IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2026-Present

- **Organization Committee**

  - IEEE FCCM 2027: Finance Chair
  - ESWEEK 2027: Ph.D. Forum & Recruitment Chair
  - ACM GLSVLSI 2026: General Co-Chair
  - ACM IGSC 2026: General Co-Chair
  - IEEE FCCM 2026: Technical Program Committee (TPC) Co-Chair
  - ACM/SIGDA FPGA 2026: Publicity and Website Chair
  - ACM/IEEE ICCAD 2026 (Track Co-Chair)
  - ACM GLSVLSI 2025: Technical Program Committee (TPC) Co-Chair
  - IEEE COINS 2025: Education Co-Chair
  - IEEE HPCA 2025: Registration Co-Chair
  - IEEE FCCM 2025: Publicity and Website Chair
  - ACM/IEEE ICCAD 2025 (Track Co-Chair)
  - IEEE IGSC 2024: Technical Program Committee (TPC) Co-Chair
  - HipChips 2024 @ MICRO 2024: HiPChips Academic Chair
  - IEEE/ACM ICCAD 2024: TPC Track Co-Chair
  - IEEE SecDev 2024: SecDev Cybersecurity Award for Practice committee
  - IEEE RAW 2024: Ph.D. Forum Chair
  - IEEE FCCM 2024: Publicity and Website Chair
  - ACM HEART 2024: Technical Program Committee (TPC) Chair
  - ACM/IEEE DAC 2023: University Demonstration Chair

- HipChips 2023 @ ISCA 2023: HiPChips Academic Chair
- IEEE/ACM NOCS 2023: Publicity Co-Chair
- IEEE FCCM 2022: Hybrid Co-Chair
- ACM/IEEE DAC 2022: University Demonstration Vice Co-Chair

- **Technical Program Committee**
  - IEEE FCCM 2026 (TPC Co-Chair)
  - ACM/IEEE ICCAD 2026 (Track Co-Chair)
  - IEEE COINS 2026 (Track Co-Chair)
  - IEEE/ACM ISCA 2026
  - ACM/IEEE Supercomputing 2026
  - ACM/SIGDA FPGA 2026
  - IEEE HPCA 2026 Industry Track
  - IEEE/ACM ICCAD 2025 (Track Co-Chair)
  - ACM GLSVLSI 2025 (TPC Co-Chair)
  - ACM/IEEE Supercomputing 2025
  - IEEE FCCM 2025
  - ACM/SIGDA FPGA 2025
  - IEEE COINS 2025 (Track Co-Chair)
  - ACM DATE 2025
  - IEEE IGSC 2024 (TPC Co-Chair)
  - IEEE/ACM ICCAD 2024 (Track Co-Chair)
  - ACM HEART 2024 (TPC Co-Chair)
  - 2024 The 1st IEEE International Workshop on LLM-Aided Design (Founding TPC)
  - ASAP 2024
  - ACM/IEEE Supercomputing 2024
  - ACM/IEEE DAC 2024
  - ACM/SIGDA FPGA 2024
  - IEEE FCCM 2024
  - ACM DATE 2024
  - IEEE/ACM ICCAD 2023
  - IEEE FCCM 2023
  - MLSYS 2023
  - ACM/IEEE DAC 2023
  - ACM/SIGDA FPGA 2023

- IEEE/CHASE CHASE 2023
- ACM DATE 2023
- IEEE/ACM ICCAD 2022
- ACM/IEEE ESWEEK 2022
- IEEE FCCM 2022
- ACM/IEEE DAC 2022
- MLSYS 2022
- ACM/IEEE DAC 2021
- IEEE SOCC 2021
- IEEE SOCC 2020
- H2RC 2020 2021 2022
- IEEE VLSI-DAT 2021

- **Conference Session Chair**
  - DAC 2026
  - FCCM 2026
  - DAC 2025
  - FCCM 2025
  - FPGA 2025
  - IGSC 2024
  - ICCAD 2024
  - ISVLSI 2024
  - DAC 2024
  - FCCM 2024
  - FPGA 2024
  - ICCAD 2023
  - DAC 2023
  - ICCAD 2022
  - MLSYS 2022
  - DAC 2022
  - FCCM 2022
  - DAC 2021
  - IEEE/ACM CHASE 2021
- **Journal Reviewer**
  - IEEE Computer Architecture Letters (CAL)

- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems(TCAD)
- ACM Transactions on Design Automation of Electronic Systems (TODAES)
- ACM Transactions on Reconfigurable Technology and Systems (TRETS)
- ACM Transactions on Architecture and Code Optimization (TACO)
- IEEE Transactions on Industrial Informatics (TII)
- IEEE Transactions on Very Large Scale Integration Systems (VLSI)
- IEEE Transactions on Computers (TC)
- IEEE Transactions on Parallel and Distributed Systems (TPDS)
- IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)
- The Computer Journal (COMPJ)

• **U.S. Government Agency Proposal Reviewer**

- National Science Foundation, 2022, 2024, 2025, 2026
- Department of Energy, 2024

• **Workshop/Special Session/Panel Organizer**

- FCCM Workshop: CASI: Custom Architectures for Scientific Instruments, 2026
- FCCM Workshop: Secure and Adaptive Computing from Genomics to Healthcare, 2025
- NSF Workshop on Algorithm-Hardware Co-design for Medical Applications, 2024
- Special Session "Design for Environmental Sustainability in Computing", Embedded Systems Week (ESWEEK), 2024

• **Subreviewer**

- FPGA 2019
- FCCM 2018, FCCM 2017
- FPL 2019, FPL 2018, FPL 2016
- HPCA 2018
- IISWC 2017
- MICRO 2017
- PACT 2016

PROFESSIONAL EXPERIENCE

Institute of Electrical and Electronics Engineers (IEEE) 2024/06 - Current
Senior Member
Only 10% of IEEE's more than 450,000 members hold this grade, which requires extensive experience, and reflects professional maturity and documented achievements of significance.

Enflame Technology 2019/08 - 2021/08
Staff Software Engineer
I worked on the biggest AI chip in China featured with 57.5mmx57.5mm package, tens of billions of transistors and hundreds of TOPs computation throughput till July 2021. My responsibilities include: 1) Pre-silicon Architecture Exploration and Modeling; 2) Post-silicon AI Software Stack Build and System Optimization; 3) Domain Specific Language for AI Accelerator including multi-level intermediate representation (MLIR)-based Compilation and Optimization. I Led a group of six-person team working on the high-performance convolution neural network library for deep learning ASIC accelerator. I also coordinated weekly progress within software teams of over 50 people and collaborated with hardware, platform, and product system teams across the whole company. **I was awarded Enflame CEO Award, and Enflame COO Award, the highest honors in the company multiple times**.

Falcon Computing Solutions (acquired by Xilinx), Los Angeles, CA 2018/06 - 2019/04
Software Engineer Intern
I worked on heterogeneous computing for genomic application by orchestrating seas of datacenter scale resources including computing (CPUs+GPU+FPGA accelerators), storage (HDD, SSD, local disk) and etc. I published MOCHA (FPGA'23) paper from my intern work at Falcon Computing Solutions.

Microsoft, Microsoft Headquarter, Redmond, WA 2017/06 - 2017/09
Software Engineer Intern Supervisor: Dr. Robert Rounthwaite
I worked in Office Machine Learning Team with my manager Robert Rounthwaite and my mentor Dr. Vincent Etter where I developed a scalable end-to-end tool to generate over 2M grammar fixed (preposition, article and etc.) sentences from Wikipedia. I also implemented a neural network model for language engine tasks.

Microsoft Research, Microsoft Headquarter, Redmond, WA 2014/06 - 2014/09
Research Engineer Intern Supervisor: Dr. Doug Burger
I work in Computer Architecture Group directed by Dr. Doug Burger in Microsoft Research, Redmond Campus. My mentor Dr. Joo-Young Kim and I worked on the image compression pipeline. I have implemented the advanced compression algorithm in C++(software reference code) and also implemented the hardware accelerator on FPGA.

CONTRIBUTING OPEN-SOURCE SOFTWARE

- **CHARM**: CHARM is an automated framework to compile matrix-multiply (MM) based applications and MM-based deep learning models onto AMD Versal adaptive compute acceleration platforms (ACAPs) hardware.
  GitHub: https://github.com/arc-research-lab/CHARM
  Publication 1: CHARM: Composing Heterogeneous Accelerators for Matrix Multiply on Versal ACAP Architecture, FPGA 2023
  Publication 2: High Performance, Low Power Matrix Multiply Design on ACAP: from Architecture, Design Challenges, and DSE Perspectives, DAC 2023

Publication 3: CHARM 2.0: Composing Heterogeneous Accelerators for Deep Learning on Versal ACAP Architecture, ACM Transactions on Reconfigurable Technology and Systems, TRETS 2024
Publication 4: EQ-ViT: Algorithm-Hardware Co-Design for End-to-End Acceleration of Real-Time Vision Transformer Inference on Versal ACAP Architecture, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, TCAD 2024

- **SSR**: SSR is an automated framework to compile end-to-end transformer-based deep learning inference models onto AMD Versal adaptive compute acceleration platforms (ACAPs) hardware.
  GitHub: https://github.com/arc-research-lab/SSR
  Publication: SSR: Spatial Sequential Hybrid Architecture for Latency Throughput Tradeoff in Transformer Acceleration, FPGA 2024
- **ARIES**: ARIES provides a Python-based frontend, an MLIR-based middle end and can target different backend devices with AMD AI Engines.
  GitHub: https://github.com/arc-research-lab/ARIES
  Publication: ARIES: An Agile MLIR-Based Compilation Flow for Reconfigurable Devices with AI Engines, FPGA 2025
- **AIM**: AIM is a compilation framework to compile arbitrary-integer arithmetic applications onto AMD Versal ACAP hardware.
  GitHub: https://github.com/arc-research-lab/AIM
  Publication: AIM: Accelerating Arbitrary-precision Integer Multiplication on Heterogeneous Reconfigurable Computing Platform Versal ACAP, ICCAD 2023
  Recorded Presentation: https://www.youtube.com/watch?v=fpUPCLNd59w
- **SCARIF**: SCARIF is a tool to estimate the embodied carbon emissions of data center servers with accelerator hardware, including GPUs, FPGAs, etc.
  GitHub: https://github.com/arc-research-lab/SCARIF
  Publication: SCARIF: Towards Carbon Modeling of Cloud Servers with Accelerators, ISVLSI 2024
  Demo: https://peipeizhou-eecs.github.io/publication/2024_isvlsi_scarif/
- **AGILE**: AGILE is the first system to enable asynchronous, GPU-initiated communication with NVMe SSDs.
  GitHub: https://github.com/arc-research-lab/AGILE
  Publication: AGILE: Lightweight and Efficient Asynchronous GPU-SSD Integration, SC 2025
  Demo: https://agile-demo.zhuopingyang.com/